Higher National Diploma in Engineering (Civil / Mechanical / Electrical / QS / BSE)

Second Year, First Semester Examination-2018

IT2102-Information Technology II A

Answer only 4 questions **Time: 2 hours**

Q1.

I. Describe the following windows on VB 6.0 environment. (2 x 4 Marks)

a) Property Window
b) Tool Box
c) Form Window
d) Code Window

II. Describe the following controls available on VB 6.0. Also specify important properties and events. (2 x 4 Marks)

a) Scroll Bar
b) List Box
c) Option Button
d) Command Button

III. Give two examples for the each of following way of event triggering. (2 x 2 Marks)

a) User interaction event triggering
b) Object event triggering

IV. Consider the following interface.

a) Explain how create buttons (1, 2, 3, 4, 5, 6, 7, 8, 9) using control array. (2 marks)

b) Write an event procedure that allow you to display a number clicked by you on the text box. Assume that the name of the text box is txtDisplay and name of the control array of the button is cmdNumber. (3 Marks)

Q2.

Describe the array variable in Visual Basic. in your explanation you have to also explain how declare array variable and how use it. (3 Marks)

I. Explain the following scope of the variables. (2 x 3 Marks)

a. Dim
b. Public
c. Private

II. The Integer is a data type to store whole number in VB 6.0. Give another four data type that can be used to store a number and describe them. (1 x 4 Marks)

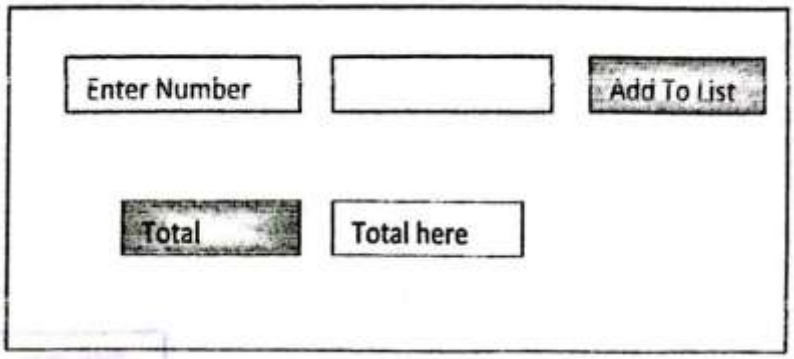
III. Write suitable variable declaration to store the following information. (1 x 4 Marks)

a. To store a student's marks. Assume mark is a whole number and maximum number of student is 50

b. To store a student's name. Assume maximum number of student is 50
c. To store a student's date of birth. Assume maximum number of student is 100
d. To store the values of following table.

10	0	-70	1000	5
40	33	22	98	12
-19	0	23	88	12
11	120	1050	18	01
23	44	19	17	24
33	22	11	99	88

V. You are requested to input a list of numbers and store in an array variable. Then you have to calculate total value of numbers you have entered. The following is the interface (VB Form).



a. Write a program (event procedure for "Add To List") to enter a list of numbers. You may assume the name of "Add To List" is cmdAddToList. (4 Marks)
b. Write a program (event procedure for "Total") to calculate total value of the list of numbers you entered. You may assume that

the name of "Total" is cmdTotal and the name of "Total here" is lblTotal. (4 Marks)

Q3.

I. Describe the following type of errors in visual basic using suitable examples. (3 x 3 Marks)

a. Syntax error
b. Run time error
c. Logical error

II. What is comment in visual basic? Explain how add comment to visual basic program. (3 Marks)

III. Consider the following program which is used to calculate the sum of list of numbers and maximum value on the list

a. Identify any four syntax errors on the above visual basic program. (4 Marks)

b. Identify any two logical errors on the above visual basic program. (2 Marks)

c. Identify at least a run time errors on the above visual basic program. (2 Marks)

d. Rewrite the above program so that the maximum value and total value of the number list should be displayed when the command button "Calculation" will be clicked. (5 Marks)

```
line 1.        ' L is array type global variable. It contain integer values
line 2.        ' N is number of element on L and N=5
line 3.
line 4.    Private function MaximumOf() as integer
line 5.            Dim M as integer, I as integer
line 6.            M= L(1)
line 7.            For I = 2 to 10
line 8.                    If L(i) < M then
line 9.                            M=L(i)
line 10.                   ENDIF
line 11.           Next I
line 12.
line 13.           MaximumOf= M
line 14.   END
line 15.
line 16.   private  function sumOf() as integer
line 17.           s as integer, I as integer
line 18.           s= 0.0
line 19.           For I = 1 to N
line 20.                   S=s + L(I)
line 21.           Next I
line 22.   end function
line 23.
line 24.  private sub Calculation_keypress()
line 25.     ' Calculation is a command button, lblResult is a label
line 26.    lblResult=  "The sum is " & sumOf() & the maximum of &  MaximumOf()
line 27.  End sub
line 28.
line 29.
```

Q4.

I. Describe the following statements in visual basic by using suitable examples. (3 x 2 Marks)

a) *For....Next Loop*

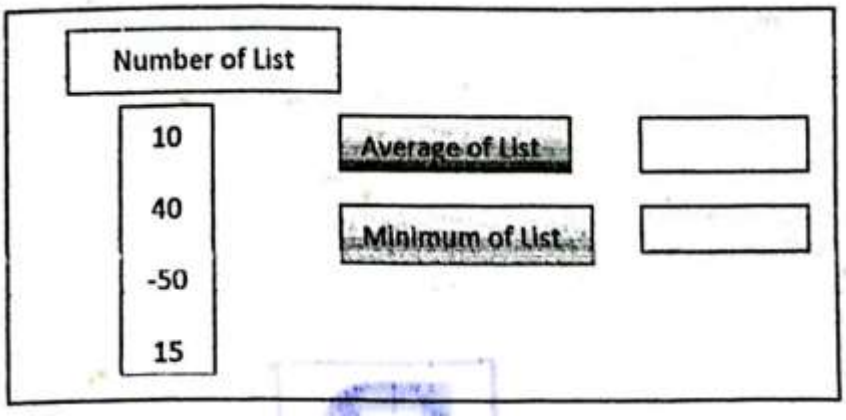
b) *Do While... Loop*

II. Describe List box control in the visual basic. (3 Marks)

III. Explain the following properties of the List Box. (4 x 1 Marks)

a. List
b. Text
c. ListCount
d. ListIndex

IV. Consider the following interface which contain a list box control, two command buttons, three labels . Assume the name of the list box is NumberList, the name of command button - "Average of List" is cmdAverage, and the name of command button - "Minimum of List" is cmbMin.



a. Explain how add the numbers 10, 40, -50, and 15 into the list - NumberList when the Form is loaded. (4 Marks)
b. Write suitable event procedure to calculate average. (4 Marks)
c. Write suitable event procedure to calculate minimum value of the list. (4 Marks)

Q5.

I. Describe the following:

a. Sub Procedure/ User define procedure. (3 Marks)
b. Function Procedure/ User Defined Function. (3 Marks)

II. Write a user define function to calculate the factorial of N, where N is positive whole number. (4 Marks)

III. Write another user define function to calculate the value of m^m where m is a whole number. (4 Marks)

IV. Design a visual basic Form to calculate the followings (5 Marks)

- Factorial of N
- The value of m^m

V. Write the necessary Code for the above the Form. (6 Marks)